

TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Delay

Creative Stereo Delay

A stereo feedback instrument where every repeat can evolve: echo, multitap, ping-pong, and tuned comb timing, driven through tape, BBD, or digital character, diffusion, frequency shifting, and grains.

Free, Sync, and Tune timing

Up to 8 taps

Grain and frequency-shift transform

99 factory presets

Version 1.0.3 · 2026-10-02

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01 Overview

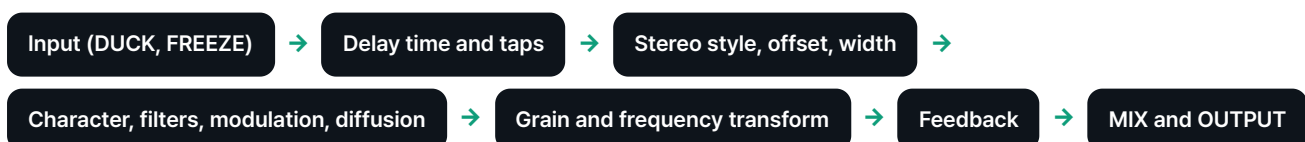
TB Delay is a creative stereo delay in which every repeat can change. Start with a clean echo, a dual delay, ping-pong, tuned timing, or tight comb textures, then drive the feedback through tape, BBD, or digital character, modulation, diffusion, frequency shifting, and granular motion.

Taps, swing, bandwidth, drive, wear, stereo movement, and ducking are shaped in one system, and built-in safeguards keep high feedback musical and controlled. COLOR and SPACE macros move many settings at once, so you can turn a preset darker or more spacious with one control.

Key features

- TIME MODE: Free (10 ms to 8 s), Sync (1/32 to 1/1 with triplets and dotted values), or Tune (tuned comb)
- 1, 2, 4, or 8 taps with SPACING and SWING
- Single, Dual, Ping Pong, and Ratio stereo styles
- Clean, Tape, BBD, and Digital character with drive, wear, filters, modulation, and blur
- Transform stage: frequency shift, grain layer, pitch, reverse, density, and spray, applied per repeat (BUILD) or once (ONCE)
- FREEZE holds the current tail; DUCK keeps the dry signal clear
- COLOR and SPACE macros
- 99 factory presets in 10 banks, plus Init and user presets

Signal flow



About this manual

This manual describes TB Delay 1.0.3 for Windows (VST3). The current macOS build is also version 1.0.3; small differences between platforms are possible.

02 Quick start

- 1 Insert TB Delay on a vocal, guitar, synth, or send. The defaults are Sync at 1/8, Ping Pong, **FEEDBACK** 48%, **MIX** 30%, **WIDTH** 130%, and **DUCK** 25%.

- 2 Choose **TIME MODE** : Free for milliseconds, Sync to follow the host tempo, or Tune for tuned comb effects.
- 3 Set **DELAY TIME** , then raise **FEEDBACK** for more repeats.
- 4 Choose **TYPE** (Clean, Tape, BBD, Digital) and shape the repeats with **LOW CUT** and **HIGH CUT** .
- 5 Set **MIX** and **DUCK** so the repeats support the source instead of covering it.
- 6 Try the COLOR and SPACE macros to push a preset darker, cleaner, tighter, or wider.

03 Interface



Captured from version 1.0.1. Since 1.0.2 the interface uses an ocean palette with a circular Delay Map and larger knobs, and since 1.0.3 the Delay Map is taller and the window can be resized from 70% to 180%. The current panel also includes DENSITY, SPRAY, the BUILD / ONCE route, and the COLOR and SPACE macros.

Area	What it does
Preset bar	Previous and next preset, the PRESETS list (FACTORY and USER), Save, Undo, and Redo.
Delay Map	Shows how the repeats of the left and right channels develop over time.
Repeat controls	TIME MODE, DELAY TIME, TIME CHANGE, FEEDBACK, TAPS, SPACING, SWING, and DUCK.
STEREO	STYLE, R OFFSET, and WIDTH, with a BYP switch.
CHARACTER / SPACE	TYPE, DISTORT, WEAR, LOW CUT, HIGH CUT, SPEED, WOBBLE, BLUR, and SPACE, with a BYP switch.

Area	What it does
TRANSFORM	BUILD / ONCE, FREQ SHIFT, GRAIN MIX, GRAIN SIZE, DENSITY, SPRAY, PITCH, and REVERSE, with a BYP switch.
Output	MIX and OUTPUT, plus the COLOR and SPACE macros and FREEZE.

04 Controls

Repeat

Control	Range · default	What it does
TIME MODE	Free, Sync, Tune · Sync	Free sets the time in milliseconds, Sync follows host-tempo note values, and Tune turns the delay into a tuned comb.
DELAY TIME	Free: 10 to 8000 ms · 375 ms; Sync: 1/32 to 1/1 · 1/8	The base delay time: milliseconds in Free, a note value in Sync (straight, T for triplet, D for dotted), and a resonant frequency in Tune.
TIME CHANGE	Crossfade, Glide · Glide	When the delay time changes, Crossfade switches cleanly and Glide slides the pitch like tape.
FEEDBACK	0 to 110% · 48%	Repeat decay and how close the delay comes to self-oscillation. Above 100% the repeats build up; safeguards keep it controlled.
TAPS	1, 2, 4, 8 · 1	Number of active rhythm taps.
SPACING	25 to 200% · 100%	Spacing between the taps.
SWING	-50 to +50% · 0%	Groove shift of the even taps.
DUCK	0 to 100% · 25%	Lowens the repeats while the dry signal plays, keeping the sound clear.
FREEZE	Off, On · Off	Smoothly blocks the input and holds the current tail.

Stereo

Control	Range · default	What it does
STYLE	Single, Dual, Ping Pong, Ratio · Ping Pong	Stereo topology of the repeats.
R OFFSET	-100 to +100% · 0%	Right-channel time ratio and comb direction.
WIDTH	0 to 200% · 130%	Stereo width of the wet signal.

Character and space

Control	Range · default	What it does
TYPE	Clean, Tape, BBD, Digital · Tape	The material of the repeats.
DISTORT	0 to 100% · 18%	Higher values make the repeats more compressed and rough.
WEAR	0 to 100% · 28%	Higher values make the repeats darker and less stable, like old equipment.
LOW CUT	20 to 2000 Hz · 120 Hz	Reduces the lows below this frequency on every repeat.
HIGH CUT	500 Hz to 20 kHz · 9 kHz	Reduces the highs above this frequency on every repeat.
SPEED	0.02 to 10 Hz · 0.25 Hz	How fast the repeats wobble.
WOBBLE	0 to 20 ms · 0.8 ms	How far the repeats wobble in time.
BLUR	0 to 100% · 12%	Higher values blur clear repeats into a diffuse, reverb-like wash.
SPACE	2 to 120 ms · 32 ms	Length and sense of space of the diffusion.

Transform

Control	Range · default	What it does
BUILD / ONCE	BUILD, ONCE · BUILD	BUILD accumulates the transformation on every repeat; ONCE applies it once at the output.
FREQ SHIFT	-2000 to +2000 Hz · 0 Hz	A signed frequency shift that accumulates on every repeat, for rising or falling spirals.
L/R SHIFT	0 to 100% · 0%	How much the left and right shifts move in opposite directions.
GRAIN MIX	0 to 100% · 0%	Balance between the normal echo and the grain layer.
GRAIN SIZE	10 to 500 ms · 90 ms	Length of each grain window.
DENSITY	1 to 8 · 4	Number of overlapping grains.
SPRAY	0 to 100% · 15%	Scatter of the grains' start positions in time.
PITCH	-24 to +24 st · 0 st	Playback pitch of the grains.
REVERSE	0 to 100% · 0%	Probability that each grain plays reversed.

Output and macros

Control	Range · default	What it does
MIX	0 to 100% · 30%	Constant-power dry/wet balance.
OUTPUT	-18 to +6 dB · 0 dB	Final output trim.

Control	Range · default	What it does
COLOR	–100 to +100% · 0%	Left gives cleaner, more open repeats; the centre is the preset as designed; right gives older, darker repeats.
SPACE	–100 to +100% · 0%	Left gives tight, static repeats; the centre is the preset as designed; right gives repeats that move more and spread wider.
BYP (per section)	Off, On · Off	STEREO bypass sets topology, offset, and width to neutral; CHARACTER / SPACE bypass skips character, filters, modulation, and diffusion; TRANSFORM bypass skips the grain and frequency transform.

The macros move several parameters together, so the individual knobs change while you turn them. Use them to explore, then fine-tune the knobs.

05 Presets

Step through presets with the arrows, or open the PRESETS list to choose a factory or user preset. Save asks for a name and stores a .tbdelay file in the User folder. The name shows Custom after you change a setting. Undo and Redo step through recent changes.

The 99 factory presets are grouped in 10 banks: Essential Echo, Grain & Freeze, Slap & Double, Rhythm & Multitap, Ping Pong & Stereo, Tape & Analog, Digital & Lo-Fi, Comb & Resonance, Spiral & Shift, and Space & Diffusion. Examples include **Clean Utility**, **Dub Throw**, **Studio Slap 80**, **Dotted Pulse Train**, **Warm Reel Echo**, **12-Bit Repeat**, **Plucked Wire**, **Gentle Lift**, **Grain Cloud**, and **Wide Ambient Wash**.

06 Recipes

Starting points for common jobs. Adjust them by ear.

Vocal throw

Repeats that answer the vocal without covering it.

TIME MODE Sync, 1/4 or 1/8 D

FEEDBACK 30 to 50%

DUCK 40 to 70%

- 1 Lower HIGH CUT to about 5 to 7 kHz so the repeats sit behind the voice.

Slapback

A short, single repeat for vocals or guitar.

Preset Studio Slap 80 or Vintage Slap 120

FEEDBACK 0 to 15%

- 1 Use Free mode and set DELAY TIME between 80 and 140 ms.

Tape echo

Warm, wobbly repeats.

TYPE	Tape
WEAR	30 to 60%
WOBBLE	0.5 to 2 ms

- 1 Lower HIGH CUT as WEAR rises to keep the repeats dark.

Spiral shimmer

Repeats that rise in pitch.

FREQ SHIFT	Positive
BUILD / ONCE	BUILD
FEEDBACK	50 to 80%

- 1 Negative FREQ SHIFT makes the spiral fall instead.

07 Tips and troubleshooting

Symptom	What to check
The repeats keep building up.	FEEDBACK is above 100%. Lower it, or use FREEZE deliberately when you want a held tail.
The delay clouds the source.	Raise DUCK, lower MIX, or lower HIGH CUT and raise LOW CUT.
Changing DELAY TIME makes pitch swoops.	TIME CHANGE is on Glide. Choose Crossfade for clean changes.
Sync timing does not follow the song.	Check the host tempo; Sync uses the host's tempo.
The sound is mono or narrow.	Check STYLE and WIDTH, and whether the STEREO section is bypassed.
Grain settings do nothing.	Raise GRAIN MIX, and check that the TRANSFORM section is not bypassed.

08 Parameter reference

Panel label	Host parameter	Range	Default
REPEAT			
TIME MODE	Time Mode	Free, Sync, Tune	Sync
DELAY TIME	Delay Time	10 to 8000 ms	375 ms
DELAY TIME (Sync)	Division	1/32 to 1/1, with T and D	1/8
TIME CHANGE	Time Change	Crossfade, Glide	Glide
FEEDBACK	Feedback	0 to 110%	48%
TAPS	Taps	1, 2, 4, 8	1
SPACING	Spacing	25 to 200%	100%
SWING	Swing	-50 to +50%	0%
DUCK	Duck	0 to 100%	25%
FREEZE	Freeze	Off, On	Off

Panel label	Host parameter	Range	Default
STEREO			
STYLE	Style	Single, Dual, Ping Pong, Ratio	Ping Pong
R OFFSET	Offset	-100 to +100%	0%
WIDTH	Width	0 to 200%	130%
BYP	Stereo Bypass	Off, On	Off
CHARACTER AND SPACE			
TYPE	Character	Clean, Tape, BBD, Digital	Tape
DISTORT	Drive	0 to 100%	18%
WEAR	Age	0 to 100%	28%
LOW CUT	Low Cut	20 to 2000 Hz	120 Hz
HIGH CUT	High Cut	500 Hz to 20 kHz	9 kHz
SPEED	Mod Rate	0.02 to 10 Hz	0.25 Hz
WOBBLE	Mod Depth	0 to 20 ms	0.8 ms
BLUR	Smear	0 to 100%	12%
SPACE	Size	2 to 120 ms	32 ms
BYP	Color Space Bypass	Off, On	Off
TRANSFORM			
BUILD / ONCE	Transform Route	Loop (BUILD), Post (ONCE)	Loop
FREQ SHIFT	Shift	-2000 to +2000 Hz	0 Hz
L/R SHIFT	Diverge	0 to 100%	0%
GRAIN MIX	Grain	0 to 100%	0%
GRAIN SIZE	Grain Size	10 to 500 ms	90 ms
DENSITY	Density	1 to 8	4
SPRAY	Spray	0 to 100%	15%
PITCH	Pitch	-24 to +24 st	0 st
REVERSE	Reverse	0 to 100%	0%
BYP	Grain Transform Bypass	Off, On	Off
OUTPUT AND MACROS			
MIX	Mix	0 to 100%	30%
OUTPUT	Output	-18 to +6 dB	0 dB
COLOR	Color	-100 to +100%	0%
SPACE (macro)	Space	-100 to +100%	0%

09 Version history

Version	Changes
1.0.3 · 2026-08-21	Fixed broken Korean text in knob tooltips. The Delay Map is taller and the control panel more compact, and the window can be resized from 70% to 180% and remembers its last size.
1.0.2 · 2026-08-19	Fixed FEEDBACK above 100% sustaining self-oscillation with no input. Changed the interface to an ocean palette with a circular Delay Map, enlarged knobs and panel titles, and tidied TIME MODE and BUILD / ONCE.
1.0.1 · 2026-08-13	Windows installer release.
1.0.0 · 2026-07-29	Initial release.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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